Data Science and AI Lab

May 2026 Term

Milestone 1 Report

Group 4

Description of problem and expected users

Problem Statement

Job search is fragmented because candidates use separate tools for resume upload, job search, role comparison, and career guidance.
This makes it difficult to identify which roles truly match their profile.

Proposed Solution

The system will use a modular AI pipeline to convert a resume into a structured candidate profile.
It will recommend suitable career roles and rank relevant job opportunities based on profile fit.

Stakeholders:

Primary: job seekers, fresh graduates, career changers, working professionals.
Secondary: recruiters and employers.

Problem Scope

Scope

The project includes resume parsing, where the uploaded resume is converted into structured information such as skills, education, experience, projects, and job titles.
It also includes candidate profile creation, so all module can use one common profile instead of taking repeated input from the user.
The system will recommend suitable career roles, discover relevant job listings, rank jobs based on profile fit.

Good to have

Provide ATS scoring, tailored resumes and personalized cover letters for selected job roles through a web-based interface.

Out of scope

The project will not verify whether a job post or company is genuine, because employer background verification is outside the current system goal.
It will not extract private contact details of recruiters or directly message recruiters
The system will not create fake skills, false experience, or unsupported achievements in resumes or cover letters.
It will also not include interview preparation and fully automatic job applications without user review.

Why this problem matter and existing solutions

Importance and cause of the problem

Why It Matters:

Job search is becoming more competitive, time-consuming, and AI-supported.
Candidates often struggle to identify which roles truly fit their skills, education, and experience.

Where It Arises:

This problem appears on online job platforms such as LinkedIn, Naukri, Indeed, Glassdoor, and company career pages.
Even with many platforms, candidates still need to manually compare jobs and judge profile fit.

Key Challenges:

- Too many job listings create confusion.
- Candidates often apply randomly without knowing the best-fit roles.
- Career guidance is usually generic.
- Candidates rarely get clear reasons for why a role or job matches their profile.

Project Importance:

This project supports candidates by creating a structured profile from their resume, recommending suitable career roles, and ranking relevant jobs based on profile fit.

Current solution available

How It Is Addressed Today

The current job search process is handled by separate tools for different tasks.

Existing solution

- Job boards like LinkedIn, Naukri, Indeed, and company career pages help users find openings.
- Career guidance tools provide general role suggestions.
- Resume builders and AI writing tools can support resume or cover letter writing.
- Academic methods study resume parsing, job matching, and career recommendation separately.

Main limitation

These tools are not connected through one structured candidate profile.
Users still have to manually compare roles, judge profile fit, and move information between different platforms.

Metrics & Success Definition

Quantitative Metrics

- **Precision, Recall, F1-Score:** Used to check resume extraction accuracy.
- **Top-K Accuracy and MRR:** Used to evaluate career role recommendations.

Qualitative Metrics:

- **LLM-as-Judge:** Rates resume tailoring, cover letters, and recommendations on relevance, clarity, personalization, and factual accuracy.
- **Human Review:** Checks whether generated content is professional, specific, and based on the candidate's real profile.

Success Definition:

- Success means accurate resume parsing, useful career recommendations and proper job discovery and matching.

Shortfall in Current Solutions

Current shortfall

Current tools are disconnected. Candidates use separate platforms for job search, resume support, and career guidance.

Key shortfall:

- Job discovery is mostly limited to major job boards.
- Users manually compare roles and profile fit.
- Career guidance is often generic.
- Job ranking is not based on one structured candidate profile.
- Feedback on why a job matches the candidate is limited..

Opportunity:

Our project connects resume parsing, career recommendation, and job matching through one shared candidate profile.
This supports candidates in finding better-fit roles through a simple modular pipeline.

Project Contribution

Main Contribution

This project is not focused on beating academic benchmarks.
It focuses on building a practical, profile-based job search support system.

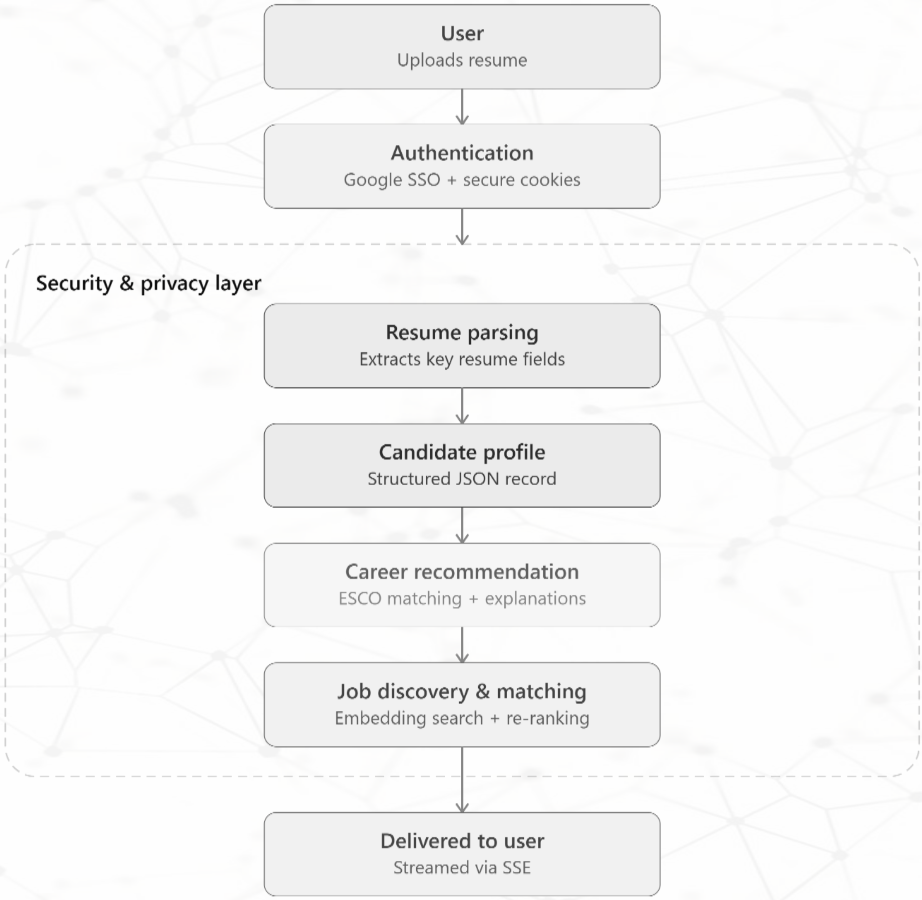
Contribution Areas:

- Combines resume parsing, candidate profile creation, career recommendation, and job matching in one workflow.
- Uses one structured candidate profile to recommend suitable roles and rank relevant jobs.
- Provides basic explanations for why a role or job matches the candidate's profile.

Proposed System Architecture / Workflow

Overall structure

Workflow: Resume upload → structured profile → career recommendation → job discovery and matching.
Authentication: Google SSO will be used, so the system does not store user passwords. Sessions will be managed using HttpOnly, Secure, and SameSite cookies.
Database: PostgreSQL will store user profiles, parsed resume fields, job postings, and evaluation results and
pgvector or ChromaDB will store embeddings for career recommendation and job matching.
Security & Privacy: Personal details such as name, email, and phone number will be masked before LLM processing and Uploaded files will be validated, and job discovery will use only allowed public pages.



Proposed System Architecture / Workflow

Parsing Module

Purpose

Converts an uploaded resume into a structured candidate profile used by other modules.

Input

Resume in PDF, DOCX, or text format.

Approach

The resume is converted into text, cleaned, and processed using an LLM to extract key fields.

Extracted Fields

Skills, education, work experience, projects, certifications, and job titles.

Output

A structured JSON candidate profile.

Evaluation

Checked using manually verified sample resumes with Precision, Recall, F1-score, and human review.

Proposed System Architecture / Workflow

Career Recommendation Module

Purpose
Recommends suitable career roles and job titles using the structured candidate profile.

Approach:

- The candidate profile is converted into a candidate context and then into an embedding using Gemini API.
- ChromaDB compares this vector with ESCO occupation vectors using cosine similarity.

Output:
Top-K recommended career roles with short explanations and missing critical skills.

Frontend Delivery:
Server-Sent Events can show progress updates such as profile analysis, ESCO matching, and explanation generation.

Evaluation:
25–30 manually curated candidate profiles will be mapped to expected ESCO occupation IDs.
Success target: at least one correct ESCO role appears in Top-K results for more than 75% of test profiles.

Proposed System Architecture / Workflow

Job Discovery and job matching

Purpose: Finds relevant job opportunities and ranks them using the candidate's structured profile, preferred role, skills, experience, location, and job type.

Process:

1. Input: Structured candidate profile from Resume Parsing and recommended roles from Career Recommendation.

2. Job Pool Creation: Job postings are collected from internal database, public datasets, saved postings, and allowed public career pages. Each posting is stored with title, company, location, skills, experience, job type, description, and posting date.

3. Semantic Matching: Candidate profiles and job descriptions are converted into embeddings using Gemini Embeddings. Cosine similarity is used through pgvector, FAISS, or ChromaDB to retrieve Top-K matching jobs.

4. Rule-Based Re-Ranking: Top matches are re-ranked using practical signals:
Skill Match 50%, Experience Fit 25%, Employment Type Match 20%, Job Freshness 5%.

5. Fallback Options: If the job pool does not return enough matches, SearXNG can search allowed public pages. If web discovery is limited, the user can manually paste a job description for matching.

6. Output: A final ranked list of job opportunities with match scores and basic reasons.

Proposed System Architecture / Workflow

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Proposed System Architecture / Workflow

Job Discovery and job matching

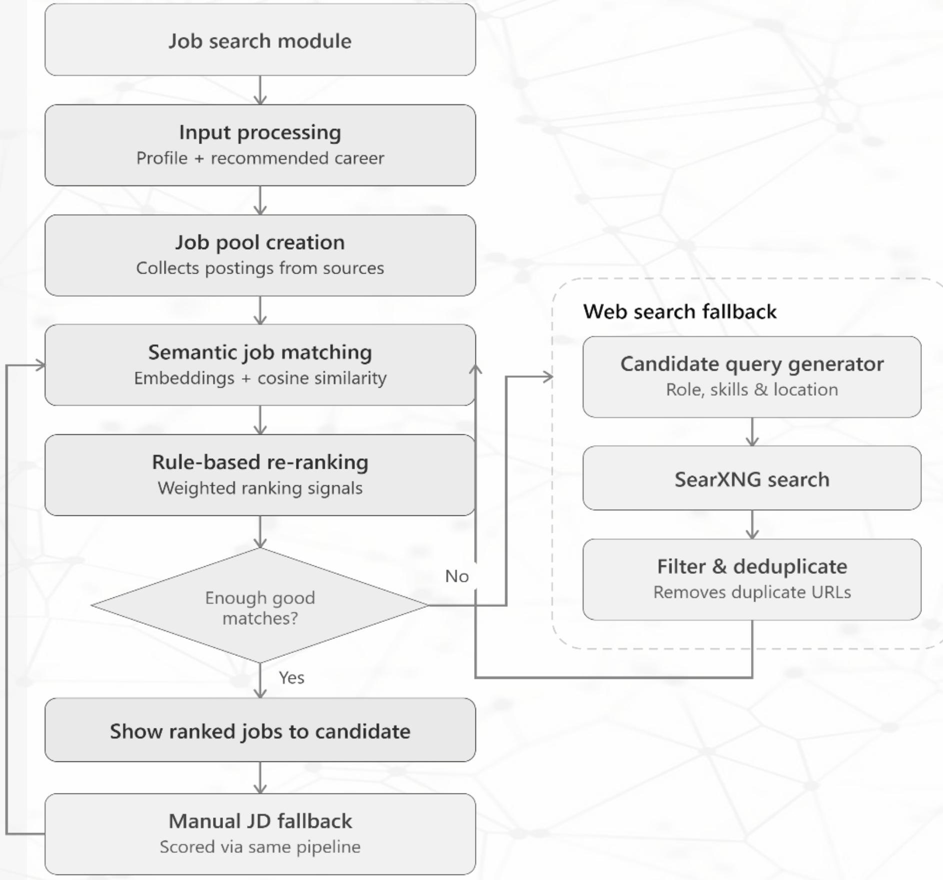
Evaluation: This module will be tested using a small set of candidate profiles and job descriptions. Possible datasets include Kaggle Candidate Job Role Dataset and Kaggle LinkedIn Job Postings Dataset.

Manual Labelling: Each resume-job pair will be labelled as: Good match, Partial match, Poor match

Ranking Metrics

- **Recall@100:** Checks whether relevant jobs are retrieved in the top 100 results.
- **NDCG@10:** Checks whether better-matching jobs appear near the top.
- **Precision@20:** Checks how many of the top 20 jobs are actually relevant.

Success Target: The module will be considered successful if relevant jobs are retrieved and better-fit jobs are ranked higher than weak matches.



Preliminary Dataset Plan

Dataset

Particulars	Expected Data Source	Purpose
Resume parsing	Public Kaggle resume datasets, open-source resume samples.	To extract skills, education, experience, projects, certifications, and job titles
Candidate profile creation	Output from Resume Parsing module	To create a structured candidate profile used by all module.
Career Recommendation	ESCO Occupations + ESCO Skills taxonomy	To map candidate skills and experience to suitable job roles
Job discovery and matching	Public job description datasets, company career pages etc.	To collect job title, company, location, required skills, experience, job type, and description
Evaluation data	Small, curated set of resumes, job descriptions, expected role suggestions etc.	To evaluate output quality and system usefulness

URL found till date

https://www.kaggle.com/datasets/ckshetty/candidate-job-role-dataset
https://www.kaggle.com/datasets/arshkon/linkedin-job-postings

Evidence of Working Approach

Purpose at This Stage:

The aim is not large-scale testing, but to show that the Generative AI pipeline works clearly on sample resumes and job descriptions

Evidence of success:

- **Resume Parsing:** Extracts skills, education, experience, projects, certifications, and job titles in structured JSON format.
  Target: F1-score ≥ 75% on a small manually checked resume set.
- **Candidate Profile Creation:** Converts extracted resume details into one clean reusable candidate profile.
  Target: 80% or more required fields filled correctly for valid resumes.
- **Career Recommendation:** Recommends suitable career roles with short explanations.
  Target: At least one suitable role appears in the Top-5 recommendations for 80% of test profiles.
- **Job Discovery:** Collects or uses job descriptions with fields like title, company, location, skills, experience, and job type.
  Target: 80% of collected jobs have key fields extracted correctly.

Evidence of Working Approach

Evidence of success:

- **Job Matching and Ranking:** Ranks jobs according to candidate-profile fit.
  Target: NDCG@10 ≥ 0.60 or Precision@10 ≥ 40% on manually labelled resume-job pairs.
- **Web Interface:** Allows users to upload a resume, view profile, see career recommendations, and view ranked jobs.
  Target: User completes the full flow without error in at least 80% of test runs.

Success Definition

The project will be successful if the core flow works end-to-end:
resume upload → structured profile → career recommendation → job discovery → ranked job matches.

Known limitations and constraints

Known limitation

- An application record is a proxy for fit: it means the user was interested, not that the job matched or that they were hired.
- Career recommendations are grounded in ESCO, which is EU-centric and slow to update, so fast-moving or region-specific roles may map poorly or be missing.
- Resume parsing accuracy may vary for highly designed or image-based resumes.
- Live job postings may expire or change.
- Web scraping is limited by website structure changes, and legal restrictions.
- Career recommendations may be biased if candidate data or taxonomy coverage is limited.
- Job ranking is a decision-support score, not a guarantee of hiring success.
- Keyword and skill overlap can't judge proficiency — matching "Python" to "Python" says nothing about depth, so a keyword-heavy resume can score close to a stronger one.

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